Software Requirements Specification (SRS)

Project: Academic Administration Software

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1. Introduction

1.1 Purpose

This SRS provides a detailed description of the Academic Administration Software (AAS). It outlines the system's purpose, scope, functional and non-functional requirements, and system interfaces. The primary goal is to automate and manage academic activities including student enrollment, faculty management, course scheduling, and examination records.

1.2 Scope

The Academic Administration Software will:

- Automate tasks related to student admissions and records.
- Manage course registrations and timetables.
- Maintain faculty details and assignments.
- Monitor attendance and grades.
- Generate reports for administration.

1.3 Definitions, Acronyms, and Abbreviations

- **AAS**: Academic Administration Software
- **SRS**: Software Requirements Specification
- **UI**: User Interface

- **DBMS**: Database Management System

1.4 References

- IEEE 830-1998 - Recommended Practice for Software Requirements Specifications
- Academic policy documents of educational institutions

1.5 Overview

The rest of this document describes system features, user interfaces, functional requirements, and other system attributes.

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2. Overall Description

2.1 Product Perspective

AAS is a standalone system that can be integrated with existing college websites or portals. It will have separate modules for students, faculty, and administrators.

2.2 Product Functions

- Student enrollment and records management
- Course registration and timetable management
- Attendance and grade tracking
- Faculty scheduling and subject allocation
- Report generation (students, courses, attendance, etc.)

2.3 User Classes and Characteristics

- **Admin**: Full control over all features
- **Faculty**: Can update grades, attendance, and manage courses

- **Students**: View personal records, register for courses, check timetables and grades

2.4 Operating Environment

- **Frontend**: Web browser
- **Backend**: Server with DBMS (MySQL/PostgreSQL), application server (Node.js/Java/Python)
- **Platform**: Windows/Linux Server

2.5 Design and Implementation Constraints

- Should comply with academic policies
- Support for English language only in UI
- Scalable for institutions with over 10,000 students

2.6 Assumptions and Dependencies

- Internet access for online usage
- Institutional data is available for import during setup

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3. Specific Requirements

3.1 Functional Requirements

FR1: Student Enrollment

- System shall allow admin to register new students and assign a unique ID.

FR2: Course Registration

- Students can view available courses and register based on eligibility.

FR3: Timetable Management

- Admin can create and manage course schedules.

FR4: Attendance Tracking

- Faculty shall mark attendance and students can view it in real-time.

FR5: Grade Entry

- Faculty can enter grades for courses, viewable by students.

FR6: Report Generation

- System shall generate academic performance, attendance, and course reports.

3.2 External Interface Requirements

3.2.1 User Interfaces

- Responsive web UI accessible on desktop and mobile
- Login portal for each user role (Admin, Faculty, Student)

3.2.2 Hardware Interfaces

- Standard institutional computer labs or user PCs

3.2.3 Software Interfaces

- Integration with institutional email systems
- API support for future integrations (e.g., payment systems)

3.2.4 Communications Interfaces

- HTTPS for secure communication

- Support for email notifications

3.3 Non-functional Requirements

Performance

- System should support concurrent usage by up to 500 users.

Security

- Role-based access control
- Password encryption and secure session management

Usability

- User-friendly design with minimal training required

Availability

- 99.5% system uptime expected

Maintainability

- Modular codebase to ease maintenance and updates

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4. Appendices

- Sample forms and UI screenshots (to be added in design phase)
- Database schema overview
- Glossary of technical terms

Project: Library Management System (LMS)

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1. Introduction

1.1 Purpose

This document specifies the software requirements for the Library Management System. The LMS will automate the process of borrowing, returning, and managing books, providing ease of access to library data for students, librarians, and administrators.

1.2 Scope

The system will:

- Manage book inventory and cataloging.
- Handle member registrations and lending transactions.
- Enable search and reservation of books.
- Provide administrative reports and overdue alerts.

1.3 Definitions, Acronyms, and Abbreviations

- **LMS**: Library Management System
- **ISBN**: International Standard Book Number
- **UI**: User Interface

1.4 References

- IEEE 830-1998 SRS Standard
- Library workflow standards of educational institutions

1.5 Overview

The remainder of this document includes detailed descriptions of the system’s functionalities, interface requirements, and constraints.

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2. Overall Description

2.1 Product Perspective

LMS is an independent web-based system that may be deployed on an institution's intranet or the internet. It will support different user roles: Librarian, Admin, and Students.

2.2 Product Functions

- Add/edit/delete book records
- Register students and manage borrowing rights
- Book reservation and renewal
- Overdue tracking and notifications
- Generate reports (borrowed books, user activity, etc.)

2.3 User Classes and Characteristics

- **Admin**: Manage users, system settings, backups
- **Librarian**: Manage books and borrowing transactions
- **Student**: View/search/reserve books, check issue history

2.4 Operating Environment

- **Client**: Any modern browser
- **Server**: Linux/Windows server
- **Database**: MySQL or PostgreSQL

- **Backend**: Java/PHP/Python

2.5 Design and Implementation Constraints

- Support for at least 50,000 book entries
- Adherence to library working hours for borrowing windows

2.6 Assumptions and Dependencies

- Internet/intranet availability
- Book catalog data imported during system setup

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3. Specific Requirements

3.1 Functional Requirements

FR1: Add/Edit/Delete Books

- Librarian can manage book records with metadata (ISBN, author, genre).

FR2: User Registration

- Admin can register and manage user accounts.

FR3: Book Search

- Students can search books by title, author, or category.

FR4: Book Issuing

- Librarian issues books with due date tracking.

FR5: Book Return and Fines

- Late returns trigger automatic fine calculations.

FR6: Reports

- Admin and Librarian can view overdue lists, borrowing history, and popular books.

3.2 External Interface Requirements

3.2.1 User Interfaces

- Clean web interface with separate dashboards for each role

3.2.2 Software Interfaces

- Integration with institutional login system (SSO, LDAP)

3.2.3 Communication Interfaces

- Email alerts for due dates and fines

3.3 Non-functional Requirements

Performance

- Support 100 concurrent users without performance lag

Security

- User authentication and role-based access

Reliability

- Data backup and restore features

Maintainability

- Well-documented code and database schema

Portability

- Deployable on cloud or local servers

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4. Appendices

- Sample UI wireframes
- Sample book catalog Excel format
- Glossary of library terms